REPUBLIC OF THE PHILIPPINES

TECHNOLOGICAL UNIVERSITY OF THE PHILIPPINES

TAGUIG CAMPUS

Bin it To Win it: An MTICS Eco-nnect IoT-Integrated Reverse Vending & Web-Based Reward System

In Partial Fulfillment of the Requirements for

SYSTEM ANALYSIS AND DESIGN

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January 13, 2026

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CHAPTER 1

THE PROBLEM AND ITS BACKGROUND

1.1 INTRODUCTION

Plastic pollution is a severe international crisis, and 400 million metric tons of plastic waste generated each year are estimated, and only 9% of it are recycled (OECD, 2022). In this scenario, the Philippines is considered to be the source of about 36 percent in the global oceanic pollution of plastic waste, equaling about 2.7 million metric tons of plastic waste every year; individually used plastic containers and sachets are the biggest perpetrators (Corrales, 2023). The cities and street life serve as micro-cities to demonstrate this dilemma, with high populations drinking bottled water, producing vast amounts of plastic rubbish. The lack of effective waste segregation system and insufficient number of recycling plants in most institutions adds to the problem of plastic bottle waste. Conventional waste management methods have failed miserably in that they are characterized by poor user involvement, recyclable material being contaminated with general waste, and lack of immediate incentive to promote behavioural change.

The emerging possibilities of Internet of Things (IoT) technologies provide a way out of these challenges through automation and an increased level of user interaction. Reverse Vending Machines (RVMs) have reached significant success in the international level; an example is the deposit -reward program in Norway with a recycling rate of beverage containers at over 90 per cent with Infinitum system at 92.3 percent in 2023 (TOMRA, 2024). However, most of the current RVM implementations are based on central database and

financial incentives, thus presenting implementation limitations on the typically limited budgets of institutions in developing nations.

In order to overcome these shortcomings, this paper presents the idea of Bin It to Win It: An MTICS Eo nect IoT -Integrated Reverse Vending / Web-Based Reward System, e.g. an integrated solution to fits the Manila Technician Institute Computer Society (MTICS). The suggested system uses a system based on an IoT-powered RVM that authenticates plastic bottles through a multi-sensor-based system and transfers Eco-Tokens to the digital wallets of students. These tokens may then be redeemed using an online reward system that would give entry to the organizational commodities, educational resources, equipment rentals, print and event attendance vouchers. This technique shows how fully eco-sustainable systems can be built, utilizing the power of IoT to reduce waste and improve engagement and administration, simultaneously, bringing real results in the form of institutional benefits to its members.

1.2 BACKGROUND OF THE STUDY

The Philippine waste management industry is faced with the acute environmental issues. According to the records of the current Secretary of the Department of Environment and Natural Resources Maria Antonia Yulo-Loyzaga, the average amount of solid waste produced in the Philippines is about 61,000 metric tonnes every day, of which 12 to 24 percent is made of plastic (DENR, 2024). PET bottles constitute a significant share of the plastic waste, as the number of such disposable packages is high. The statistics on consumption of the product across the globe also reveal the scale of the issue: the global sales of plastic bottles have already reached one million units per minute, which is 1.44 billion bottles per day, but the problems with recycling of such materials is still insufficient due to

the insufficient infrastructure of segregation and ineffective approaches to recovery (Climate Change Commission, 2024).

Empirical research has shown that a mere presence of recycling bins does not imply proper use. Lack of incentives and prompt feedback systems is one of the key barriers to the highest recycling rates and prevention of contamination (The Recycling Partnership, 2023). This value action gap, or the discrepancy between attitude and behaviour regarding the environment is a specific problem with young people, who often sacrifice ecological values in the name of convenience. The traditional recycling frameworks based on individual consciousness only have not been able to maintain high recycling rates. Additionally, the cash-based deposit-refund schemes have challenges to implementation in most developing situations, including the complexity of accounting facilities and the high frequency of cash processing. These problems highlight the need to have incentive structures that will directly compensate the participants and have few administrative complexities.

By integrating the Internet-of-Things (IoT) technology into the waste management systems, many of the flaws of the traditional strategies are eliminated due to its automation and validation in real time. Smart garbage systems that are sensor-driven and connected devices have proved to be relatively successful in boosting the efficiency of collection and lowering the environmental impact. However, the current reverse-vending-machine (RVM) implementations are usually relying on centralized cloud databases and constant internet connection, which makes them challenging to apply in developing environments where internet connection is not reliable and cloud subscriptions mean ongoing expenses. In turn, the architecture of autonomous, locally controlled RVM systems becomes one of the key elements to facilitate recycling automation in the resource-limited settings.

The effectiveness of token-based incentive systems to elicit recycling behaviour has been testified. The research demonstrates that recyclable waste could increase by more than 229% in case tokenised reward mechanisms were implemented in communities (Jieyu et al., 2021). Within the framework of this institutional context, the current research aims to find a solution to a radically topical environmental issue in terms of its technological feasibility and environmental sustainability.

1.3 STATEMENT OF THE PROBLEM

Other organizations such as the Manila Technician Institute Computer Society (MTICS) which consists of students pursuing the Information Technology and Computer Science course of the Manila Technician Institute are very instrumental in managing the services of academic support and other activities of the students in the IT department. Nevertheless, all the activities of such organizations were performed with semi-digital procedures such as paper documents and personal emails.

In the present scenario, there are challenges associated with the system(s) of MTICS when it comes to validating members, monitoring sanctions, reservation of equipment, monitoring attendance during events and rewarding members. There is also no centralized system connecting the member information of the students in the organization and the respective services, rewards etc that are offered on those services, etc. The officers are then confronted with the challenge of ensuring validity of activity of the members, sanctioning, and making authentic reports ready to be audited and accredited.

Moreover, they have the environmental programs such as recycling programs that are promoted at a departmental level; however, they are not integrated into reward system and engagement systems of the organisation easily. The lack of an automated IoT-based and

web-based platform also restricts the capacity of the organization in incentivizing the students to engage, monitor their contribution to the environment and in fact connect those to the real benefits in the services of the MTC.

Hence, it is reasonable to have a centralized, role-based and transparent system, which will combine organizational management, resource services, event participation, and reward ecosystem, backed by IoT-enabled recycling in case of the occurrence of such issues. When fixed, the above-mentioned shortcomings will be a part of improving the efficiency of operation, accountability, student engagement, and data-driven decisions in MTICS.

1.3.1 MAIN PROBLEM

The Manila Technician Institute Computer society (MTICS) does not have a single, IoT based, and web-based organizational management ecosystem which can centralize records of members, impose role-based access control, automate organisational services and sanction processes, integrate a recycling-based reward system, and transparent analytics. This lack of presence leads to inefficiencies of operations, lack of accountability and disconnection of data and involvement of students in Information Technology and Computer Science division Manila Technician Institute.

The key issue is that it is impossible to find one digital platform that will facilitate all administrative, operational, and engagement-related tasks of MTICS. The extensive use of hard copy documents, spread sheets and various other digital instruments currently in organizational processes has posed the risk of inconsistency in data, slow approvals, ineffective implementation of policies, and inefficient reporting. This necessitates a central system, which can sustain both short term organizational processes, and also long term institutional governance and match efforts of MTICS with sustainability and student engagement goals.

1.3.2 SUB-PROBLEMS

1. Role-Based Access Control

At present, the controls do not exist in the form of a proper security role-based access control system to enable and grant student and officer/faculty accounts to access data, services and administrative functions of the organization without any controls or improperly patrolled.

2. Automated Sanction Management

The system currently in use by MTICS lacks a tool that will automatically recency and monitor a student in terms of sanctions applied in the organizational policies of particular penalties of non-attendance during prearranged events. This will create a lack of consistency in its execution and allow students who have outstanding sanctions to proceed with enjoying rewards and organizational services.

3. Resource, Facility, and Service Automation

The MTICS operate on manual and disjointed processes in handling equipment loans, facility reservations and printing services which contributes to the lack of real-time visibility, delayed approvals, increasing burden on administration and frequent inefficiencies in such services.

4. Event Management and Attendance Monitoring

There is no centralized digital platform that processes organizational events in the MTICS and so the events are created manually hence resulting into inefficientness in tracking attendance or errors in event records. This slows down the issuance of certificates to participants of the events.

5. IoT-Integrated Recycling and Eco-Token Allocation

In the present state, the secure processing of the data that is generated by IoT-enabled reverse vending machines is not a mechanism implemented in the MTICS, thus, linked to the

distribution of Eco-Tokens to the members and/or the monitoring of the recycle process in a way that does not allow any tampering.

6. Digital Reward Marketplace and Inventory Control

How does the system enable to create a safe digital marketplace, where students can earn MTICS approved rewards with use of Eco-Tokens, as well as a system of sufficient inventory management, confirmation of transactions validity and limited release of items by relevant officers.

7. Analytics, Reporting, and Organizational Transparency

Currently, the Manila Technician Institute Computer Society (MTICS) does not have a safe online marketplace where Eco-Tokens can be redeemed, and this aspect is an issue that the organization is struggling with as it relates to how inventory, user validation of the transactions and the controlled distributed rewards by the official.

1.4 OBJECTIVES OF THE STUDY

This study is driven by the goal to shift toward smarter campus practices - blending IoT tools with real-world motivation strategies. Instead of relying on outdated trash pickup routines, attention turns to building something more responsive through “Bin it To Win it,” a two-part digital setup made for the MTICS team. Automation checks each recycled PET bottle, triggering rewards via an online hub that students access directly. This approach tests whether small-scale tech solutions can boost eco-awareness without heavy investment. Alongside improving recycling habits, it opens new paths for handling everyday department needs efficiently. The effort unfolds around clear aims, both broad and detailed in scope

1.4.1 GENERAL OBJECTIVES

This project focuses on building an Internet-connected recycling station called "Bin it To Win it," combined with a web tool named "MTICS Eco-nnect." Its purpose is to form one connected system that uses sensors to check returned PET bottles automatically. Instead of just discarding plastics, students receive electronic rewards known as "Eco-Tokens" for taking part. These tokens act as points earned through proper disposal behavior. Connected directly to a school-specific online hub, the setup allows users to trade their points for campus benefits like Wi-Fi time or printer use. Over time, such integration supports smarter handling of institutional resources while promoting eco-friendly habits across the organization.

1.4.2 SPECIFIC OBJECTIVES

This study specifically aims to:

1. **To design and fabricate the "Bin it To Win it" Reverse Vending Machine (Hardware)**
    - By integrating a Mass-Density Verification System (Load Cell + HX711) and an Inductive Proximity Sensor to autonomously distinguish valid PET bottles (15g–45g) from contaminants such as glass, metal cans, and liquid-filled containers.
    - To implement an electromechanical sorting mechanism driven by high-torque servo motors that physically segregates accepted materials into the storage bin while rejecting invalid items, thereby ensuring high-purity waste collection at the source.
2. **To develop the "MTICS Eco-nnect" Web Portal (Software)**

- By creating a locally hosted, responsive web application (via ESP32) that replaces manual logbooks with a Digital Membership Database for student profiling and organization management.
- To engineer a Real-Time Synchronization Protocol that links the physical kiosk to the web server, ensuring that "Eco-Tokens" are credited to the student's digital wallet instantly upon a successful bottle drop.

3. **To implement a centralized digital redemption ecosystem (System Integration)**
    - By programming a Token-Based Transaction System where accumulated credits can be exchanged for specific department services.
    - Specifically, to automate the provisioning of Internet Access (via MikroTik Captive Portal integration), Printing Services (via Job Queue Management), and Equipment Borrowing (via Inventory Reservation), thereby streamlining the operational workflow of the MTICS officers.
4. **To evaluate the technical performance of the system (Testing)**
    - By measuring the Detection Accuracy (Sensitivity and Specificity) of the sensor array in identifying valid vs. invalid bottles.
    - By testing the System Latency, specifically the time delay between the physical bottle drop and the reflection of points in the user's web account, to ensure a seamless user experience.
5. **To assess the user acceptability and usability of the system (Evaluation)**
    - By conducting a survey among MTICS students and officers using the ISO 25010 Software Quality Standards.
    - To determine the system's effectiveness in terms of Functional Suitability, Usability, and Reliability, ensuring it meets the operational needs of the organization and the behavioral motivations of the students.

1.5 SIGNIFICANCE OF THE STUDY

The formulation of the MTICS Eco-nnect system has significant advantages to the different stakeholders as it seeks to solve the two problems of effective waste management and digital accessibility in the department.

- **To the Environment** The project also improves Source Segregation that is a major contributor to success in the recycling process. The project ensures that the PET bottles recovered in the community will not contain any contaminants, including liquid or non-plastic objects, that will diminish the recyclable value of the PET bottles. This project also ensures that the plastic waste does not seep out to the community and the drainage system.

- **In the case of the MTICS Department,** it is a cost-efficient automated solution that meets the compliance with the environmental requirements of the department like the RA 9003, the Ecological Solid Waste Management Act. It saves the student officers and maintenance staff a lot of work since it saves on the sorting processes that are carried out manually. This enhances the brand of the department as a pioneer of Smart Campus by showing a viable application of the IoT technology to address the actual ecological challenges.

- **To the Students** The system makes the process of recycling more rewarding and interactive instead of a passive task. The system uses positive reinforcement to promote long-term environmental stewardship behavior by providing real economic rewards, i.e., free internet access, printing credits or merchandise. It narrows the digital divide on students who otherwise have access to very limited mobile data

resources, and ensures that students have access to the connectivity they require to succeed.

- **To Future Researchers** This study offers a basis on the construction of IoT based incentive setups alongside web-connected devices. It describes a strategy that is comprehensively expanded by demonstrating how physical recycling activities are connected with online reward systems. The same machine may be used for recyclables or even events in the neighborhood. The design can be changed without failure.

1.6 SCOPE AND DELIMITATION

Scope of the Study

The main idea of the study is how to design, develop, and introduce the Bin it To Win it system a composite of a smart recycling bin with the help of the IoT and the MTICS Eco-nnect site. One of them is renewing the way the resources are managed at the department with digital incentives on the students and automation of trash collection.

The technical and operational constraints of the study are the following functionalities:

- **The implementation of IoT Hardware (The Kiosk):**
    - **Sensor-Based Checking:** There is a weight-based check, that is executed initially - with a load cell - to permit items between 15g and 45g, which are of typical empty size PET bottles. Pieces of metal in the trigger cause rejection as detected by an inductive sensor operating simultaneously. The combination of these two checks does not overlap to provide precision.

- **Automated Sorting:** The system has a gating mechanism that is servo controlled and physically separates acceptable bottles into an internal bin and rejects invalid ones back to the user.

- **Web-Based Management System (The Portal):**
  - **Membership and Access Control:** There are safe logins based on user roles, and only the members of the MTICS can get access. Member information is now not stored on paper but through a well structured digital medium. The advantages include the fact that it is easier to update when statuses change. The security is enhanced by individualizing account controls. It is also quicker to monitor who is meant to be where without people having to sign-in as they do physically. There is also more accuracy in the system as it eliminates human error in entering. Every individual receives a distinct profile associated with his or her position. The outdated files and spreadsheets are substituted with digital checks.
  - **Digital Economy (Eco-Tokens):** Digital economy is powered by eco-tokens, which are stored on a virtual wallet. These credits are received by students via live API signals from the kiosk. While printing or claiming these items, tokens are deducted automatically. The transactions are updated real-time and balances are updated accordingly. Every transaction is automated and does not require human intervention.
  - **Resource Management:** A software application manages the reservations of shared equipment - i.e. HDMI cords or computer stations utilized by employees. Another component of the same installation is followed up by tracking stored rewards.

- **Connectivity & Integration:**
    - The combination of a Raspberry Pi is used as the local web server, which runs the portal as well as its corresponding database. Speaking more precisely of live data exchange, the system is connected with the physical kiosk and the online access point of students via Wi-Fi. The feedback is instant, and every time recycling is run, it is updated, and this is achieved through constant syncing of devices.

Delimitations of the Study To make the project a viable endeavor within the academic period, the following limitations and exclusions are applicable:

- This equipment can only work with PET plastic bottles. It is designed in a manner that makes it compatible only with that type of material.
    - **Other Recycling Exclusion:** Glass bottles are not accepted and even aluminum cans are not accepted. Tetra packs cannot be processed in this case. Waste paper is also rejected.
    - **Sanitation Procedures:** The machine rejects bottles containing any significant quantity of liquid or unused material - more than 50 grams. This measure prevents the damages of internal electronics and ensures the maintenance of hygiene standards.
- **Hardware Limitations:**
    - **No Compaction:** Space is not compressed by compression without the use of built-in crushing machines such as industrial compactors or grinders. Consequently, the number of bottles that can fit directly depends on their original shape and size.

- **Indoor Deployment:** Only one unit is placed in a hallway or office. But it is unable to confront fields outside on which the sun or rain strikes.

- **Platform Constraints:**
  - **No Mobile Application:** This system is based on a web platform, which can be viewed using any common phone browser such as Chrome or Safari. Only these browser environments are accessible on mobile devices. Native Android or iOS apps are not even within the scope of the project.
  - **Department-Level Scope:** This platform was initially designed to work on its own, without the involvement of MTICS. Although there are wider institutional means, it does not share data with either the OSA or the office of the Registrar.
- **Network Dependency:** There is only one working connection (wireless or wired) that is needed to be fully operational. Bottles can be entered in the machine without internet connection, but until a connection with the central systems is restored, no rewards are possible. As long as there is no local connection made, validation stops in its tracks. Internal hardware has not yet supported cellular service (like 4G or similar standards).

CHAPTER 2

REVIEW OF RELATED LITERATURE

1.1 FOREIGN LITERATURE

In a bid to combat the shortage of useful recycling mechanisms in the developing world, such as Pakistan, where the traditional Reverse Vending Machines (RVM) are either too costly and complicated to establish, Zia et al. (2022) designed a more affordable and automatic RVM that employs the deep learning technique to scan plastic bottles with precision. The system uses an ESP32 microcontroller to control a set of custom internal gears and arms that sort the bottles by size once detected by sensors and a camera with the help of a machine learning model to differentiate between large and small bottles and provide WiFi connectivity to notify users of the rewards and monitor bin status. Though there was some initial problems with the system in the form of scanning accuracy because of lighting and angle of the bottle, these problems are now solved and the system is more precise, strong and smaller in size compared to the earlier versions making it an effective tool to use in the outdoor world despite some minor noise in the motor and the sensors.

To solve the challenge of getting households to adopt recycling as a habit because of the lack of motivation, Gibovic and Bikfalvi (2021) organized a study in a small town in Spain to find out whether raffle-like incentives would increase the rate of recycling. The researchers made use of a digital platform in which 150 households received tokens using a QR code in bags which allowed the researchers to compare the effectiveness of small, guaranteed rewards with the promise of larger rewards using a lottery scheme. It was positive and 1,000 families registered and the quality of waste was significantly increased as the

contamination was reduced, the study concluded that the lottery scheme was more appealing than the small discounts, close to 60-percent of the users used their tokens on weekly cash games.

Rao and others (2020) concentrated on the fact that plastic waste is damaging the environment even with the modern technologies, and it presents a system to promote the use of rewards to encourage people to collect and recycle their waste. The researchers put this to the test by deploying smart bins that included WiFi, weight scales, and fill-sensors enabling the user to scan a QR code before placing their waste and get 2 points by placing 100 grams of plastic in the smart bin throughout 12 weeks. The research noted that there was a progressive growth in the volume of the recycled plastic as well as the activity of the users with the statistics demonstrating that the rates of collection grew significantly hence showing that the appeal of rewards was successful in encouraging people to recycle.

Ortega and Ferrer (2025) also believed that inadequate recycling rates at home and in offices are usually a result of ignorance about how to sort trash, and hence created a smart trashcan, iTrash, which is an incentive-based easy-to-sort trashcan. During five days in a university environment, iTrash tested the use of cameras and sensors to identify objects, color-coded lights that would direct the user to the right bin, and would offer digital rewards to sort into the correct bin. The outcomes have shown that the behavior has greatly improved: whereas regular trashcan users sorted waste correctly only 47% of the time, the iTrash system scored 82% waste recognition, which proves that automated feedback and rewards are effective in enhancing sorting behavior.

The next concern, which Mustafa (2025) covered, was the problem of the cyclical nature of recycling programs over time since the constant reward system leads to the waning interest of the participants eventually. The research suggested a conceptual model, which was

not applied to the real world but through 10,000 simulations which applied a token-based reward system in which the value of tokens varies according to the demand in the market and is tied to carbon credits to motivate corporations to participate in the program. The researcher was intentionally trying to keep the user motivation high by introducing dynamic rewards instead of fixed ones because in this way, a market-driven approach would have ensured that recycling would become more and more appealing, though less predictable in the short term with the uncertainty of the token values.

1.2 FOREIGN STUDIES

One of the researches reviewed by Zia et al. (2022) was focused on the topic of increased volumes of plastic bottle waste, which lies in the creation of a low-cost incentive-based Reverse Vending Machine (RVM) to manage plastic waste. The study underlined the idea that traditional options of recycling can either be computationally high or may not be robust enough to be used over a long term by people. The researchers developed an RVM to eliminate all these restrictions, incorporating lightweight image processing, sensor technology, and a reward based user application to make the population participate. The framework relied on a MobileNet deep learning model that was trained using transfer learning to classify PET bottles with a testing error of 99.2, and the model compares to a much smaller size than other current classification methods. The results of deploying two units to a university campus indicated that the system was able to collect more than 650 kilograms of plastic waste in six months because of incentives that motivated customers into recycling their objects and encouraged automated systems. The research supports the current study, because it confirms that a sensor-built reverse vending machine with digital reward system can be implemented and it is cost effective, contributing to the fact that the suggested

system called Bin It to Win It has a potential to enhance recycling behavior and waste sorting in the institutional context.

In a study by Gibovic and Bikfalvi (2021), the effectiveness of incentive-based recycling schemes was studied with the introduction of a virtual token of a reward referred to as RECICLOS in Catalonia, Spain. Due to a growing number of European recycling objectives of plastic, the study was driven by the necessity to persuade households to participate by gamifying and incentivizing it through digital tokenization, electronic registration, and by a web-based enrollment tool. The pilot project that took six weeks included 1,053 families, which was 10 percent of the target population, and it also came to implement new reward systems, which included raffles, lotteries, tax reductions, and charities. The results indicated that there was a significant change of increasing the recycling behavior with an improvement in active participation in recycling behavior by 60 per cent to 80 per cent and also significant improvement in proper disposing of waste. The paper has highlighted that clear, interactive, and non-financial incentive programs have a great impact on the recycling and maintain user interest. The current investigation lends credibility to the current study since it was able to prove that a token-supported reward system and online platform are capable of influencing the utilization of recycling behaviors, proving the usefulness of the suggested Bin It to Win It system in encouraging long-term involvement and open-efficiency reward distribution in an institutional context.

An article by Wankmuller et al. (2023) investigated the application of blockchain-based tokenization as an incentive tool to enhance the collection of plastic bottles in the enterprise. Due to the drawbacks of traditional incentive plans, the study discussed the fault system grounded on tokens, which are to be given to end-customers as a reward of good recycling of plastic bottles. The study tested different token-based incentives, i.e., donations,

lotteries, gamification strategies, etc. through an empirical pilot study in partnership with the industry partners, including Coca-Cola HBC. The results indicated that the technology of incentives based on blockchains and tokens is technically viable and efficient in raising the rate of plastic bottle collection, decreasing the use of virgin materials, and enhancing the effectiveness of recycling as a whole. It is worth noting that the findings revealed token based incentives having social purpose to be the most effective in influencing consumer participation in it. Through this, the current study can be seen as highly beneficial to the current research because the evidence provided is that digital token reward systems have the potential to increase recycling activity and visibility and chicken, as well as the suggestion of the current research, that the Eco-Token system, namely Bin It to Win It, should be implemented at the institutional level to encourage sustainable recycling behavior.

A research on machine learning algorithms and machine learning-based smart waste management and recycling by Ginting and Apnena (2024) was done using an arduino-based waste sorting system. The research covered the problem of rising amount of waste in contemporary urban settings and the significance of automated systems in enhancing the efficiency and sustainability of waste management. The authors used Support Vector machine (SVM), and Neural Network (NN) to categorize various types of waste and place them on the right colored bins through two machine learning models. The SVM model attained an accuracy rate of 92 percent with the Neural Network model recording a 96 percent accuracy rate during testing. Both models yielded good results with ideal perfection, recall, and F1-score, meaning that the system was able to detect and classify waste without any misclassification. The significance of using advanced technologies, namely, IoT, hyperspectral imaging, lidar, and other machine learning algorithms, towards enhancing the accuracy and scalability of waste sorting in further research were also highlighted in the

study. This study is appropriate in the design of the smart waste management systems capable of promoting recycling and environmental sustainability. (Ginting & Apnena, 2024).

Filho, De Brito, Paiva, Klemensov, and De Oliveira (2024) investigated the combination of the IoT and smart bin technologies to enhance the process of recycling waste collection process. In their investigation, they stopped at automating the collection of waste by measuring the fill-up of bins with sensors that gave the data on the real-time conditions of the waste collectors to determine the most efficient routes and times placed the waste collectors in the field. To provide the reliability and security of the system, the data collected was sent safely and stored, on a blockchain platform, which increases the level of transparency and eliminates the possibilities of manipulation of the data received. The study highlighted the importance of the inclusion of security protocols like encrypted data and access control in order to maintain data privacy and integrity. Comprehensively, the paper has found that the integration of IoT, smart bins, and blockchain can provide substantial efficiency and save money and help facilitate a more sustainable recycling culture, as it offers accountability and better waste management tools.

1.3 LOCAL LITERATURE

The Extended Producer Responsibility (EPR) Act of 2022 (Republic Act No. 11898) represents a significant shift in the Philippine legislative space with regards to waste recovery. This statute commits large enterprises to implementing circular economy by recycling a significant proportion of their plastic packaging waste with incremental goals of reaching 20% in 2023 and 80% by 2028. According to PCX Markets (2025) although the law places a burden on producers, it also creates a need for efficient recovery mechanisms that have the abilities to track and verify the diversion of waste. The literature suggests that in

order for this policy to be successful, recovery infrastructure must go beyond traditional landfills and move towards "upstream" solutions to capture waste at the source. As a result, the proponent's study is validated, as the "Bin it To Win it" system provides the exactly verifiable, tech-enabled recovery mechanism that institutions need to comply with national EPR mandates and fill the currently available void in accessible infrastructure for compliance.

In terms of the national strategy, the World Bank and the DENR's "Roadmap for the Management of Plastic Waste" (2024) highlight the urgent need to move from the "linear economy" (take-make-dispose) that the Philippines currently does, which mismanages 80% of plastics. The roadmap suggests that the country loses a huge amount of economic value as it fails to recycle materials, especially single-use plastics and PET bottles. It highlights a serious missing link in the local collection systems that can sort out the garbage with enough purity for the recycling market. This scholarship shows the need of the proponent's device, which uses a mass-density verification system to assure high purity collection (effectively segregated liquids and non-plastics). Accordingly, it is a direct response to the call in the roadmap for better material quality to provide a foundation for a viable domestic recycling industry.

Support for the introduction of Deposit Return Systems (DRS) has already gained momentum amongst industry leaders too. A workshop held by the Samahan sa Pilipinas ng mga Industriang Kimika (SPIK) in 2025 inquired into the possibility of nationwide implementation of DRS to boost circularity. Industry experts at the forum argued that incentive-based schemes are the most successful way of significantly raising beverage container collection rates and pointed to global success stories. Nevertheless, as noted by the

literature, the present state of the Philippine context is that widespread automated machinery, in the form of Reverse Vending Machines (RVMs), is non-existent necessary to implement DRS on a large scale, and hence relies on manual collection. The "Bin it To Win it" project solves this technological gap by proposing an alternative RVM based on cost-effective and locally fabricated techniques that can operationalize the DRS concept within an educational institution and act as a microcosm for wider nationalization.

With regard to behavioral reinforcement, recent efforts of the Metro Manila Development Authority (MMDA) and private sector partners (2025) have seen the success of "trash - to - cash" or reward models. In a program initiated in collaboration with the Jollibee Group, the people of the community were incentivized to trade in segregated plastics for gift certificates and food items, leading to increased community engagement. Whilst this endeavour was successful in demonstrating that Filipinos are responding positively to economic incentives for recycling, it was reported that the process was still very manual, relying on physical weighing and logbooks. This offers an obvious gap in terms of operational response which is being addressed by the proponent's study, by automating validation and reward through IoT and digital wallet ("Eco - Tokens"), the study eliminates the administrative burden of manual incentive programs and maintains the high levels of engagement that are seen in the MMDA's manual incentive programs.

The need for "Plastic Smart Cities" in Metro Manila, as being pushed by WWF-Philippines (2023) highlights the role of adopting localized solutions for urban waste management. The initiative calls for 30% reduction of plastic leakage by enabling barangays and local institutions to have more effective waste management logistics. However, existing literature on Smart Cities in the Philippines tend to focus on macro-"solutions", such as garbage truck tracking systems or sanitary landfills, while giving less importance on

"point-of-disposal" technologies that deal with individual users. The proponent's study adds meaningful context to this Smart City story with the introduction of a granular, user-centric solution—a smart bin—that digitizes the recycling experience at the campus level, presenting a bottom-up approach to the smart city framework that is underrepresented in discussions in Singapore.

1.4 LOCAL STUDIES

In the article titled Basupera: A Feasibility Study on Reverse Vending Machines in the Philippines, Castillo et al. (2019) used a financial and operational feasibility assessment to determine the feasibility of implementing an RVM system to redirect the segregation process toward consumers through incentives in Metro Manila, where the plastic garbage problem is on the rise. With the help of an elaborate feasibility framework and cost-benefit analysis in Quezon City, the researchers discovered that the city produces about 12 million kilograms of PET waste every year, which allows confirming a sustainable supply to make RVMs commercially feasible in high-traffic sites such as universities, particularly when advertising fee is involved. The researchers came to the conclusion that RVMs are highly socially profitable in terms of creating long life periods at landfills and it is an important economic and environmental value to justify the Bin It to Win It project. This confirms the basic assumption of the proposed system that waste recovery with the help of automation and incentives are effective and effective and it presents statistical evidence that academic institutions provide enough volumes to support such technology, which the supporters are trying to adapt with cheaper locally-developed equipment.

As part of a research on the global waste crisis, Baribad et al. (2024) created the Plastic2Fantastic reverse vending machine which is powered by the use of aluminum as its

material and aims to encourage people to recycle their materials. The system had a Raspberry Pi 3B with a coin hopper and a complete set of sensors such as ultrasonic, capacitive and inductive proximity sensors so that the system could detect the material correctly. The highest functionality and acceptance of the prototype was confirmed by a performance testing, as it proved that multi sensor fusion is a competent foundation of bottle recognition. Nevertheless, though Baribad et al. proved that this sensor setup can be useful, they used a Raspberry Pi and mechanical coin dispenser, which was more expensive and required more maintenance. This technical background underlies the current project under consideration, Bin It to Win It, which, however, overcomes the mentioned shortcomings by introducing a more affordable ESP32 microcontroller and a scaleable and maintenance-free digital reward system instead of a mechanical dispenser.

Gito, Sarmiento, and Espinosa (2024) concentrated on the combination of technology and conservation by creating a so-called Wireless Internet-Connected Incentive System at Aemilianum College Inc. The authors of the study created a unit, where the automated mechanism of bottle separation and user-friendly interface were tested in a strict compliance with ISO 25010 industry standards. The system scored a weighted mean of 4.06, which is much more than what can be expected, which indicates that internet connectivity is a powerful influencer in student recycling acts. This research is the main source of confirmation of the business model of Wi-Fi to Trash applied in the proposed project. Nevertheless, there is a major difference in data integration, where Gito system functions as the independent program, whereas the presented project performs further this idea by integrating the reverse vending machine into the wider scope of the MTICS Ecosystem. The proposed system will transform the work into an Eco-Token by linking the data of the recycling to the organizational profile of a student and allowing extending the advantage of

the reward beyond the mere connectivity to the administrative perks, such as attendance at the events and clearances.

The work by De Leon and Battung (2021) aimed at updating waste management in the City of Ilagan by creating an IoT-based product named E-Recycle Bin to be implemented in the Barangay Material Recovery Facilities. The researchers also used Rapid Application Development (RAD) to design and develop a prototype that used a Raspberry Pi and an Arduino Mega along with certain sensors to identify and weigh waste loads such as load, ultrasonic, and IR sensors. The system used a GSM module to provide cloud-based analytics and a thermal printer to print hard copies to give as incentives. The analysis conducted by experts according to ISO 25010: 2015 criteria demonstrated that the weighted average of the indicators was 4.91, which confirms that the system is of high quality and can be used as a waste mitigation instrument. Although this research presents important confirmation of the application of load cells to measuring waste and effectiveness of reward systems, its application of GSM modules and thermal printers will result in repeatable expenses of data and paper material. The Bin It to Win It system that will be proposed will solve these financial limitations through the use of a local Intranet and a paperless digital token system. This pivot can remove the continuous operational expenses, and this is a more cost effective idea that suits non-profit student organizations.

Marzan, Olarte, Pugeda and Gumabay (2025) dealt with the operational inefficiencies that were tormenting student organizations in Saint Paul University Philippines due to the use of manual records and the lack of data in a centralized location. Based on a descriptive-developmental design and the Agile Scrum model, the researchers created a centralized platform that focused on removing the errors in tracking and simplifying the

administration. The software was tested in terms of ISO/IEC 25010 standards, where it is determined to be in compliance with ISO/IEC 25010 standards to a significant level in terms of functionality, usability and reliability, which contributed to a great extent to the accuracy and accountability of the functioning of the software. This paper critically justifies the importance of having a single web portal to curb the manual inefficiencies in student leadership. Nevertheless, whereas the system suggested by Marzan et al. revolves around the majority of the inactive administrative data, the suggested MTICS Ecosystem goes one step further and incorporates the real-time IoT and QR code interaction. This development makes the system not merely a record keeping mechanism but a dynamic platform, which actively interacts with the students providing automated attendance and environmental incentives.

1.5 IPO DIAGRAM

Table 1. IPO Diagram for Bin it To Win it: An MTICS Eco-nnect IoT-Integrated Reverse Vending & Web-Based Reward System

INPUT	PROCESS	OUTPUT
Knowledge Requirements: 1. Knowledge in IoT concepts and smart waste management 2. Knowledge in Reverse Vending Machine (RVM) operations 3. Knowledge in web-based systems and database management 4. Knowledge in reward and token-based systems Developer's Knowledge: 1. Basic knowledge in ESP32 microcontroller 2. Knowledge in sensors (Load Cell, HX711, Inductive Sensor) 3. Knowledge in HTML, CSS, PHP, and MySQL 4. Knowledge in system analysis and design 5. Software Requirements 1. Arduino IDE (ESP32 programming) 2. HTML, CSS, PHP (Web portal development) 3. MySQL Database (Eco-Token and user records) 4. Web server Hardware Requirements:	**START** ↓ **Idealization** • **Identification of plastic waste problems within the MTICS department** • **Conceptualization of an IoT-based Reverse Vending Machine combined with a web-based reward system** • **Idea to motivate students through Eco-Tokens instead of cash incentives** ↓ TITLE DEFENSE ↓ **Background of the Study and Statement of the Problem** • **Discussion of plastic pollution issues and ineffective recycling practices** • **Identification of MTICS operational problems** • **Establishment of the need for an IoT-integrated recycling and reward platform** ↓	Bin it To Win it: An IoT-Integrated Reverse Vending Machine and Web-Based Reward System

1. ESP32 Microcontroller
2. Load Cell with HX711 Module
3. Inductive Proximity Sensor
4. Servo Motors (sorting mechanism)
5. Power Supply and Wiring Components
6. Ultrasonic Sensor
7. Reverse Vending Machine Enclosure
8. Laptop/Desktop

Processor

1. ESP32 Dual-Core Microprocessor
2. Local Server Processor

RAM

1. ESP32 Internal RAM
2. Server RAM for database and web application hosting

Data Gathering

- **Review of related literature and studies**
- **Collection of system requirements**
- **Identification of hardware requirements**
- **Identification of software requirements**

System Development
Hardware Development

- **Assembly of Reverse Vending Machine components**
- **Integration of Load Cell + HX711 + Proximity Inductive Sensor + Ultrasonic Sensor**
- **Servo-controlled sorting mechanism**

Software Development

- **Development of MTICS Eco-nnect web portal**
- **Implementation of user login, and Eco-Token system**
- **Integration of IoT data transmission between ESP32 and web server**

System Implementation

- **Deployment of the Reverse Vending Machine within the MTICS environment**
- **Connection of hardware to the local intranet/web server**
- **Real-time testing of bottle insertion, validation, and Eco-Token crediting**
- **User testing by students and officers**

END

1.6 DEFINITION OF TERMS

Deposit Return System (DRS) - A system of incentives whereby consumers are offered money back when they return containers of the beverages. Digital tokens are substituted with cash in this research to promote the reuse of plastic bottles.

Eco-Tokens - Students receive a virtual currency as a reward of successfully recycling bottles. Such tokens are stored in digital wallets and can be redeemed to services like internet access, printing or to use equipment.

ESP32 - A low-power microcontroller that is cheap and is needed to control sensor feedback and coordinate data exchange within the reverse vending machine, which will be a cost-effective alternative to the Raspberry Pi.

Extended Producer Responsibility (EPR) - Cites Republic Act No. 11898, a policy that holds producers liable with regard to the post-consumer waste. This is the law that justifies the use of automated waste recovery systems in the study.

HX711 - An analogue to digital converter with high precision, which is the interface of the microcontroller and the load cell enabling the system to detect the weight of the inserted bottles accurately.

Inductive Proximity Sensor - An object detector that functions on the principle of non-contact sensor. It sifts through cans of metals or foil to make sure that all plastic items are taken in during the recycling exercise.

ISO 25010 - A global standard on the quality of software used in this research to determine how the functional suitability, usability, and reliability of the software were through surveys conducted on students and officers.

Job Queue Management - An automated printing workflow computerized system. It enables students to place documents in a print queue by use of Eco- Tokens and hence simplifies the service delivery and tracking resources.

Load Cell - A force-to-electrical signal transducer, which is the main sensor to ensure that the weight of the bottle is within the acceptable weight range of 15g to 45g.

Local Intranet - An intranet to run the system without the use of the internet. It minimizes data expenses and provides 24/7 and safe networking that is restricted to members and university personnel.

Mass-Density Check System - A load cell and HX711 were used to weigh items as a hardware validation feature. It does not take in any heavy items or those with liquid, but takes only empty bottles of PET (15-45 g).

Polyethylene Terephthalate (PET) - Another example of thermoplastic polymer in liquid containers. In this paper, the research is limited to the acceptance of empty, clean PET bottles so that they can be recycled.

Reverse Vending Machine (RVM) - A kiosk that can be used automatically and enable people to have their empty beverage containers returned to get a reward. It makes use of sensors to c

Role-Based Access Control (RBAC) - A security system which limits access to the system depending on the user roles. It separates student and officer accounts in order to provide authorized access to certain services and administrative functions.

Servo Motor - A physical sorting mechanism consisting of a rotary actuator. It works based on very accurate angular control to place valid bottles into storage or the invalid items back into the hands of the user.

System Latency - The time lag between the actual physical deposit of the bottle and the comparable reflection of the points into the user web account which was to be measured such that the user experience is responsive.

Value Action Gap - The disillusionment between the environmental beliefs and the real actions of an individual. The solution to this gap is a system that uses digital rewards to align

the daily activities to ecological ideals, confirm the authenticity of the bottles and separate useful objects and garbage.

Developmental Research Design - This is a methodological tool that is being used to conceive, execute, and support the IoT-enabled recycling device and its corresponding web portal by systematic prototyping and technical analysis.

Agile Iterative Design Method - A logical model of development adopted by the study team to build both the hardware and software components through the continuous production of design, development and improvement cycles.

Internet of Things (IoT) - A technology platform that can be used to automate the verification of the bottle and send live data between the physical kiosk and the central web server by the use of sensors and networked devices.

Quantitative Experimental Testing - A controlled experimental protocol is a protocol designed to measure and quantify objective system performance, e.g. sensor precision and response latency.

System Performance Metrics - Official technical requirements such as the detection accuracy and system throughput are used to determine the effectiveness and feasibility of the overall solution.

Data Synchronization - refers to the real time connection between the physical kiosk and the web server, which means that Eco Tokens can be added to the electronic wallet of a student as quickly as possible after a bottle deposit has been authenticated.

Unit Testing - A preliminary test stage where individual hardware and software units are empirically tested to be operating correctly, by elaborate performance logging.

MySQL - A relational database management system which handles the secure storage, retrieval and management of the user records, transaction history and Eco-Token balances.

Integration Testing - This is a comprehensive evaluation phase, which assures the integrated functionality of the different hardware and software components, including the ESP32 microcontroller and the web server, in a single system architecture.

User Authentication - A security system that will involve the use of role based access controls which will be used to determine the credentials of the students or officers before they are granted access to the data and services of the organization.

CHAPTER 3

RESEARCH DESIGN AND METHODOLOGY

3.1 RESEARCH METHOD

This paper employed a Developmental Research Design, particularly, prototyping and technical testing of a hardware-software system. The authors used the Agile Iterative design method to develop, implement, and realize the IoT-based recycle machine and the web portal.

The study has used Quantitative Experimental Testing, instead of traditional social surveys used to measure objective performance of the system. This was to be done through a series of controlled experiments to document the accuracy of the machine in sensor recognition, the time response of the sort mechanism and the accuracy of the data synchronization with the server. The study determined the technical feasibility and efficiency of the proposed solution by addressing the System Performance Metrics instead of the subjective opinion of the user.

3.2. RESPONDENTS

The Bin it To Win it: An MTICS Eco-nnect IoT-Integrated Reverse Vending and Web-Based Reward System was created to use an IoT-enabled kiosk and web platform to facilitate the process of recycling and organizational service management. The system is mainly being answered by the active student members and officers of the Manila Technician institute Computer society at the Technological university of the Philippines Taguig Campus, as the main users and the administrators of the platform. The primary task of the project is to incorporate a localized digital ecosystem that would allow real-time validation of recycling, allocation of Eco-Tokens, resource management, and access to web-based services, and the

researchers could test the usability, functionality, and effectiveness of the given system in the real context of the functioning of the organization

3.3. RESEARCH INSTRUMENT

In order to provide the thorough assessment of the system Bin It to Win It, the researchers used Technical Testing Checklist and Performance Log, which is aimed at the objective evaluation of the functionality and efficiency of the hardware and software components. This tool was also used as the main data collection tool in the Unit Testing and the Integration Testing stage involving the substitution of traditional survey questionnaires with empirical performance measurements. The checklist was designed in such a way that it would validate the key behaviours of the system, such as whether the Inductive Proximity Sensor was accurate in measuring whether it was dealing with a metal object or a plastic one and whether the Load Cell was accurate in measuring the rejection of a bottle with a liquid inside it.

The Software Functionality Log was also part of the instrument to determine reliability of the MTICS Web Portal. The Pass/Fail criteria were used in this section to verify the principal characteristics of the use of the ESP32 microcontroller and the MySQL database including the ability to synchronize the data in real-time between them, the ability to provide user authentication, and the creation of unique Eco-Token vouchers successfully. In order to provide reliability of the results, the authors performed 30 trials per hardware test case, documenting the success rate and response latency of each trial. The information obtained through these logs formed the quantitative foundation of the review of the second criterion of the Functional Suitability and Reliability of the system against the goals of the project.

3.4. DESIGN PROCEDURE

The design process of Bin it To Win it: An MTICS Eco-nnect IoT-Integrated Reverse Vending and Web-Based Reward System is systematically planned and oriented in such a way that the system can meet the demands of both students and administrators and provide a convenient and efficient experience to the users. Several major aspects in the design process are flowcharts and diagrams which are a visualization of the functionality of the system, the user interface as well as data flow. These visualizations will help to learn about the processes and interactions of the system, ensuring the well-arranged system that suits the objectives of the project.

A comprehensive review of the requirements is the first step in the design process that aims at identifying the issues facing MTICS in its management of organizational affairs and environmental programs. This analysis entails management of information related to the members, automation of tracking sanctions, recording of equipment borrowing, attendance control at the event, and introduction of IoT-based recycling and a token-based reward system. The resulting system will aim at availing a centralized platform through which operational efficiency, accountability, and student engagement will increase and which will foster sustainable recycling behaviour.

The design process includes both software-based and hardware-based elements: the Bin it To Win it Reverse Vending Machine with sensor-based validation of the choice and the MTICS Eco-nnect web portal with role-based access control and real-time integration of IoT data thus forming an ecosystem that connects physical recycling process with online rewards and organizational services.

3.4.1. Program Flowchart

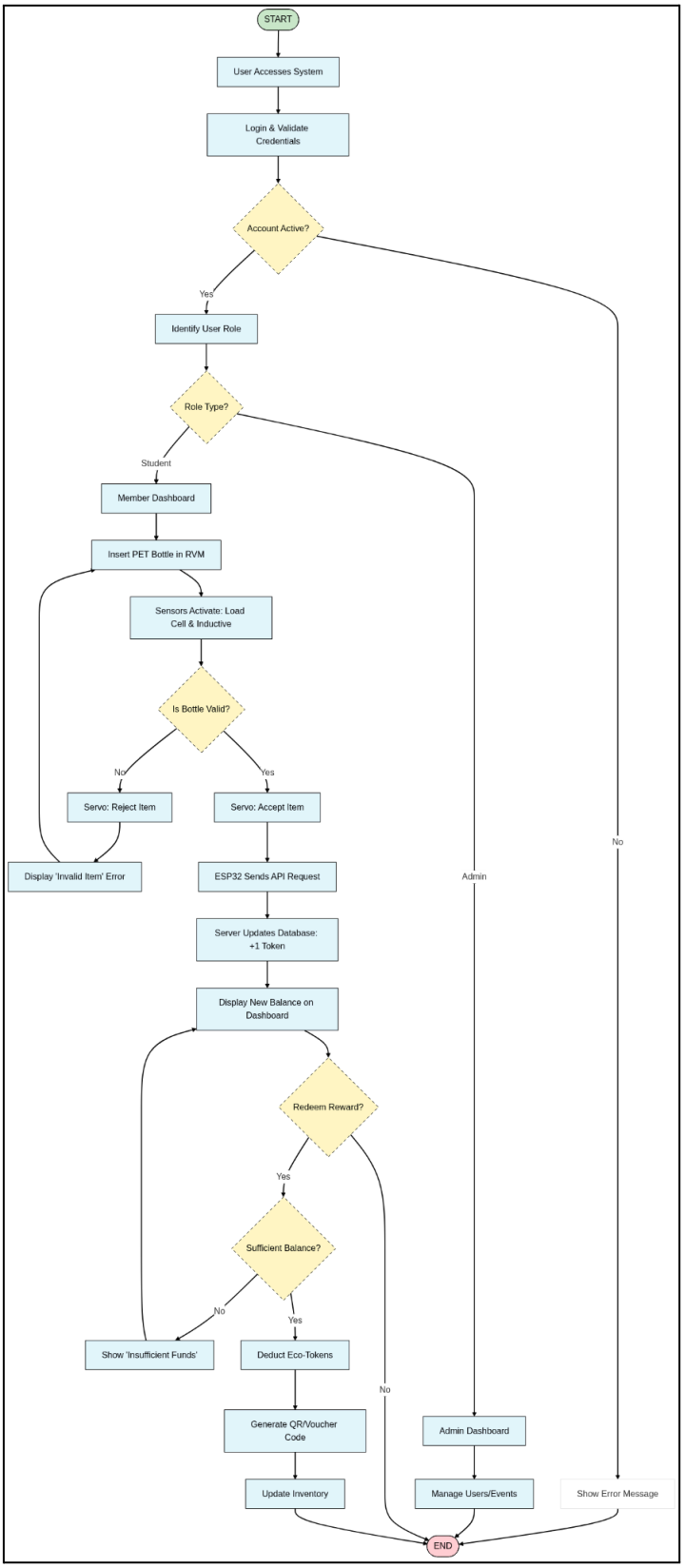

Figure 1. Program Flowchart of Bin it To Win it: An MTICS Eco-nnect IoT-Integrated Reverse Vending & Web-Based Reward System

3.4.2. System Flowchart

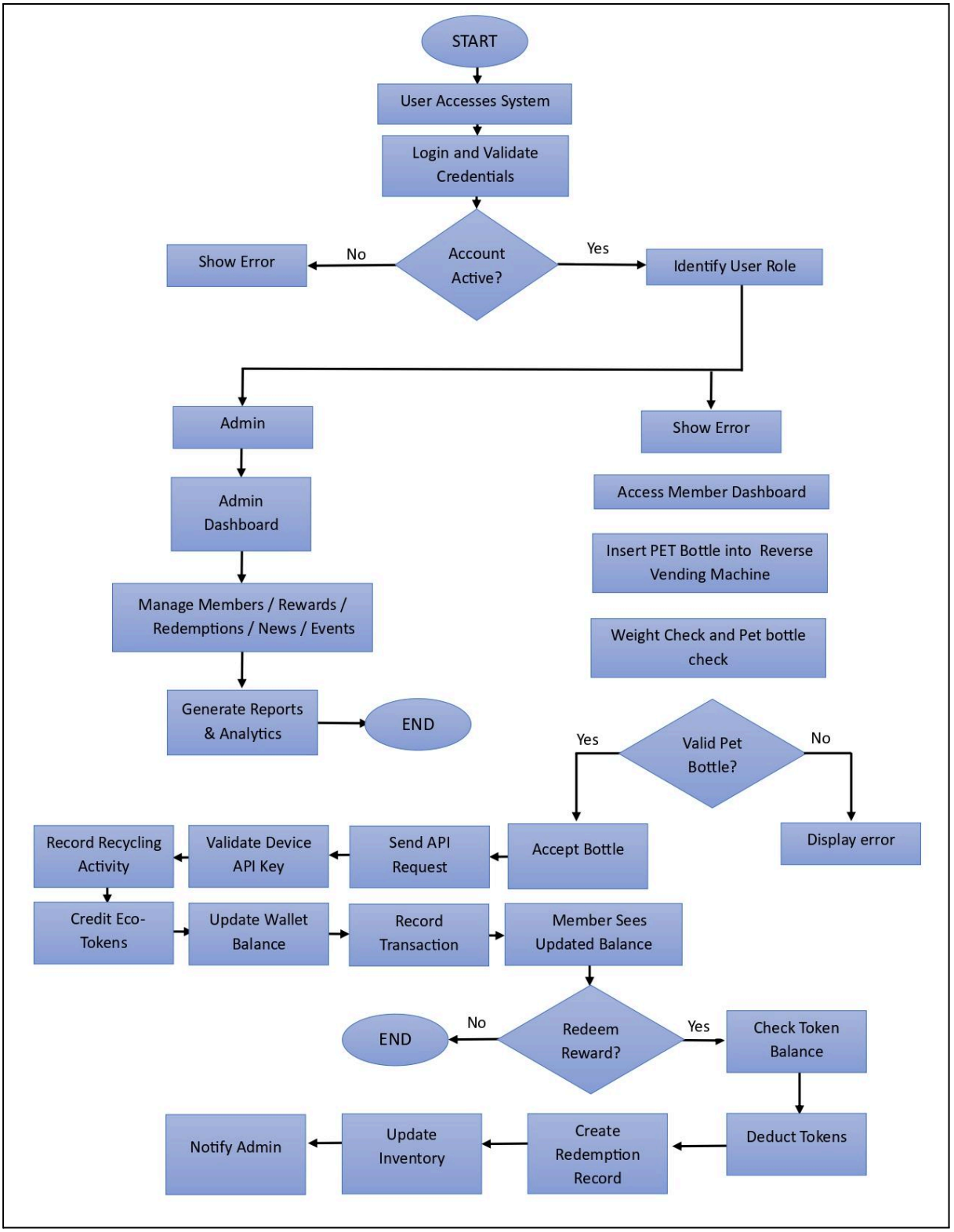

Figure 2. System Flowchart of Bin it To Win it: An MTICS Eco-nnect IoT-Integrated Reverse Vending & Web-Based Reward System

3.4.3. Context Flow Diagram

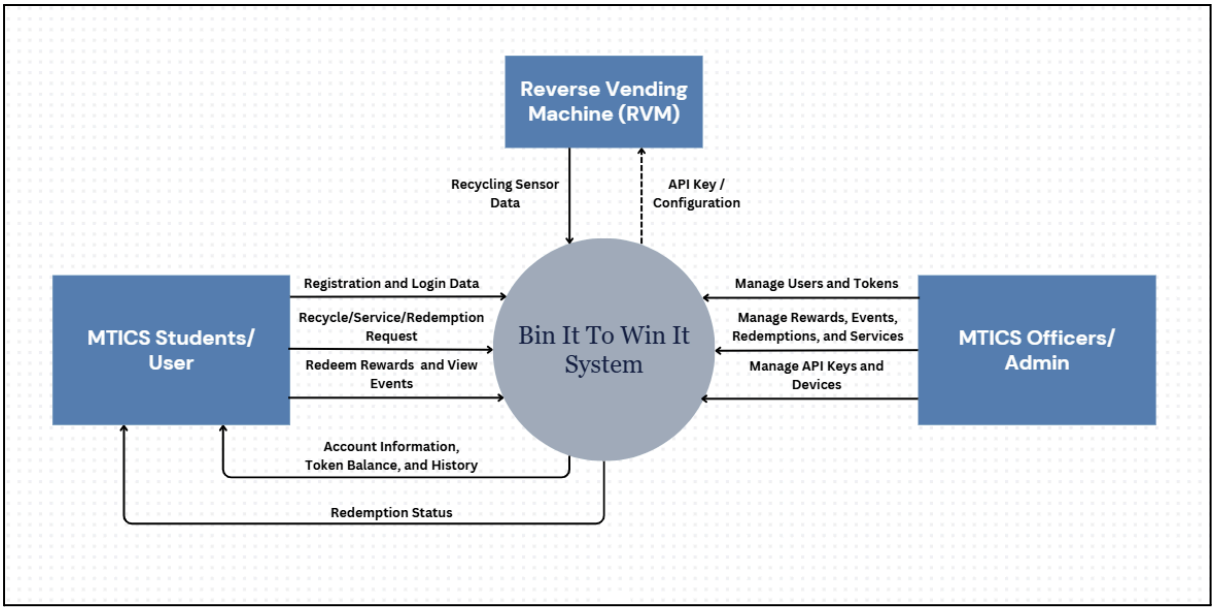

Figure 3. Context Flow Diagram of Bin it To Win it: An MTICS Eco-nnect IoT-Integrated Reverse Vending & Web-Based Reward System

The Bin It To Win It System Context Flow Diagram represents the main interactions of the Bin It To Win It System between the MTICS Students/Users, the MTICS Officers/Admins and the Reverse Vending machine (RVM). MTICS Students can also create accounts, log in, and see the listing of the events, place a recycling or redemption order, and collect the reward. The system, in its turn, provides them with the account details, the balance of tokens, the histories of transactions and the redemption statuses. On the other hand, the officers of MTICS are the administrators of the platform that control users, tokens, rewards, events, redemptions, services, and API keys. The system also ensures that it keeps constant contact with the RVM hardware, and gets the necessary recycling sensor data, as well as providing the machine with the necessary API settings. Overall, the chart shows the flow of information and interactions and, therefore, makes it clear that the system will have automated processes of recycling and reward.

3.4.4. Data Flow Diagram

3.4.4.1. Data Flow Diagram Level 0

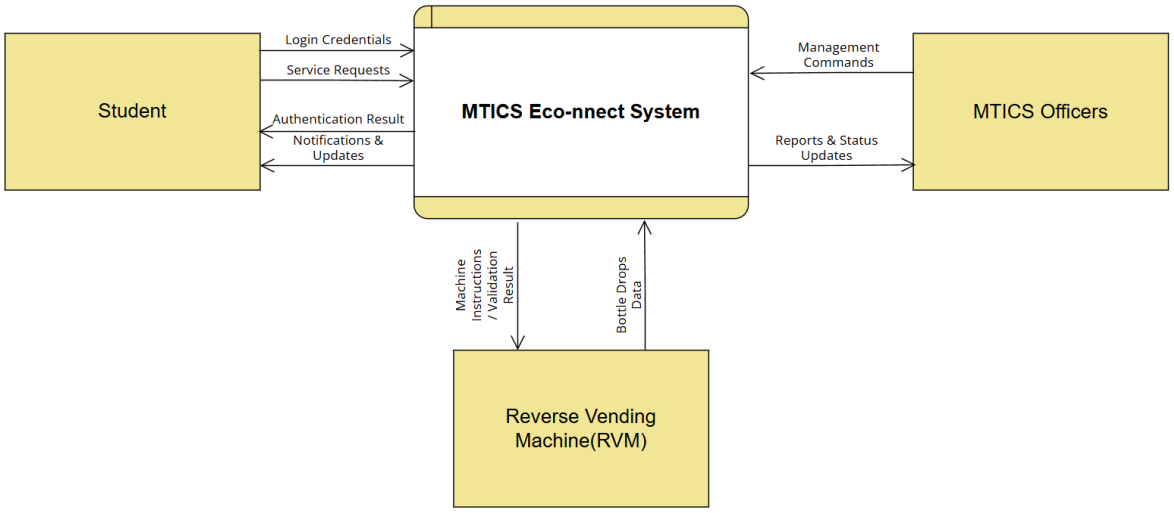

Figure 4. Data Flow Diagram Level 0 of Bin it To Win it: An MTICS Eco-nnect IoT-Integrated Reverse Vending & Web-Based Reward System

The Level 0 Data Flow Diagram of the MTICS Eco-nnect System is a high-level description of the interaction between the system and external parties and the way the system processes user requests. The students are able to log-in, request drop off of the bottles, redeem the rewards and book reservations, with the Reverse Vending Machine (RVM) transmitting bottle information and validation feedback to the system. The MTICS Admin is a system that controls rewards, inventory and produces reports. These activities are processed by the system and updated in the corresponding records and tokens balances, confirmations, and notification are sent back to the students, which allows running the eco-friendly transactions and rewards with ease.

3.4.4.2. Data Flow Diagram Level 1

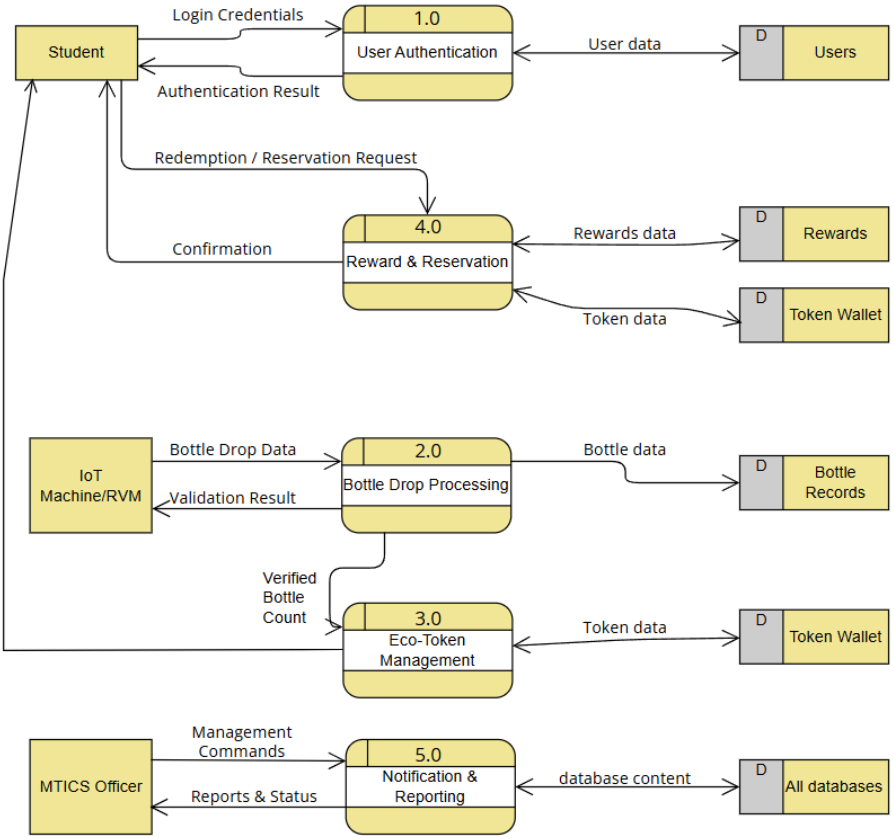

Figure 5. Data Flow Diagram Level 1 of Bin it To Win it: An MTICS Eco-nnect IoT-Integrated Reverse Vending & Web-Based Reward System

The Level-1 Data Flow Diagram of the MTICS Eco-nnect System divides the overall system into five major subprocesses, which include interactions with the external parties and internal data repositories. The User Authentication process is where students provide their login details that are validated against the User Database providing authentication feedback. Bottle drop, reward redemption and reservation requests are submitted by students and processed by Bottle Drop Processing and Reward & Reservation Management in case they are part of eco-friendly activities. The bottle information is entered into the Bottle Drop Records and the eco-tokens updated in the Eco-Token Wallet by the Eco-Token Management process, where students are issued with updated balances of eco-tokens. In the meantime, the

MTICS Officer initiates the management commands and retrieves the notification and reports through the Notification and Reporting process, where all the necessary data stores can be accessed. The IoT Bottle Machine enables the access to real-time bottle drop and the validation outcome to be delivered, this way the system would track the recycling activity and compensation effectively and the confirmation and notification would be returned to students to keep them engaged and responsible.

3.4.4.3. Data Flow Diagram Level 2

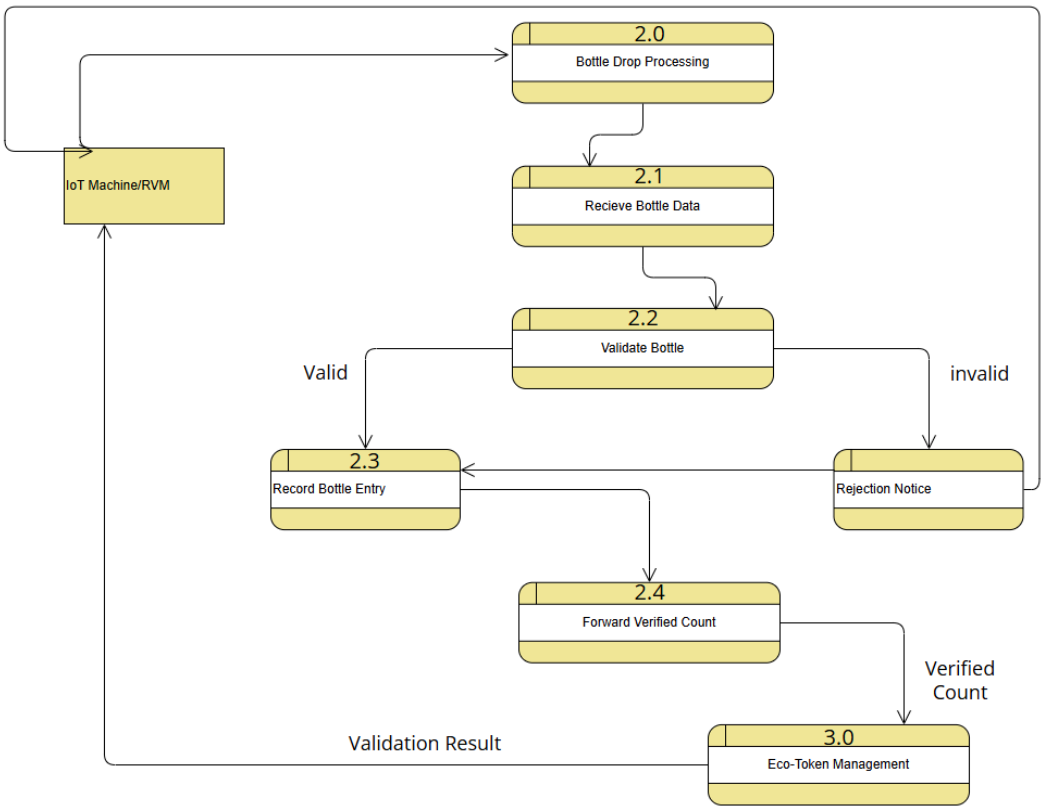

Figure 6. Data Flow Diagram Level 2 of Bin it To Win it: An MTICS Eco-nnect IoT-Integrated Reverse Vending & Web-Based Reward System

Level-2 Data Flow Diagram 2.0 Bottle Drop Processing breaks down the parent process in Level-1 into four subprocesses to indicate in detail how it interacts with the IoT Bottle Machine and the Bottle Drop Records database. The first one is 2.1 Receive Bottle

Data which receives raw bottles drop data on the IoT Machine then forwards it to 2.2 Validate Bottle which validates against bad bottle types, duplicates and correct drop procedures. In the case of invalid bottle, a rejection notice is returned to the IoT Machine; in the case of valid bottle, the data has been sent to the 2.3 Record Bottle Entry, where the confirmed bottle information is stored in the D2 Bottle Drop Records. Lastly, 2.4 Forward Verified Count uses the updated token count to 3.0 Eco-Token Management to update the token count and sends a validation response to the IoT Machine to complete the flow. This Level-2 diagram not only maintains all Level-1 inputs and output, but demonstrates decision processing at the validation step, and distincts internal processing, data store interactions and external entity communications, maintaining a reasonable flow balance and traceability.

3.5. ENTITY RELATIONSHIP DIAGRAM

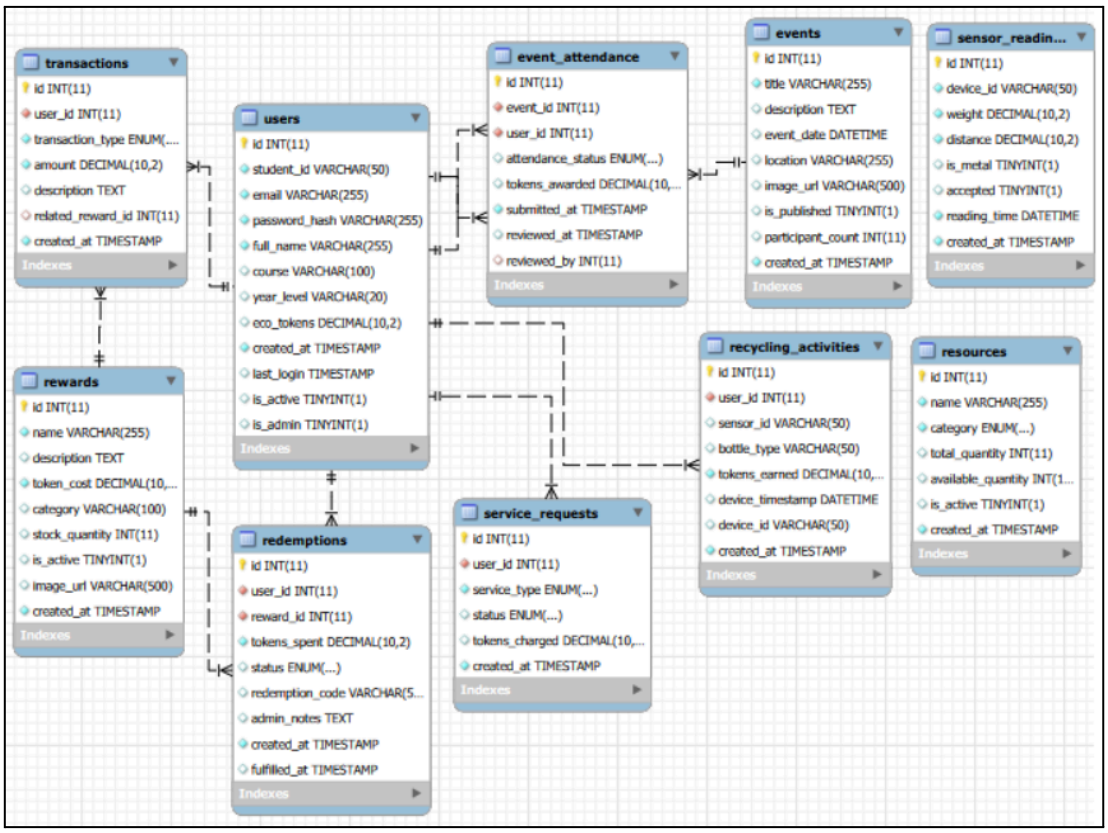

Figure 7. Context Flow Diagram of Bin it To Win it: An MTICS Eco-nnect IoT-Integrated Reverse Vending & Web-Based Reward System

3.6. USE CASE DIAGRAM

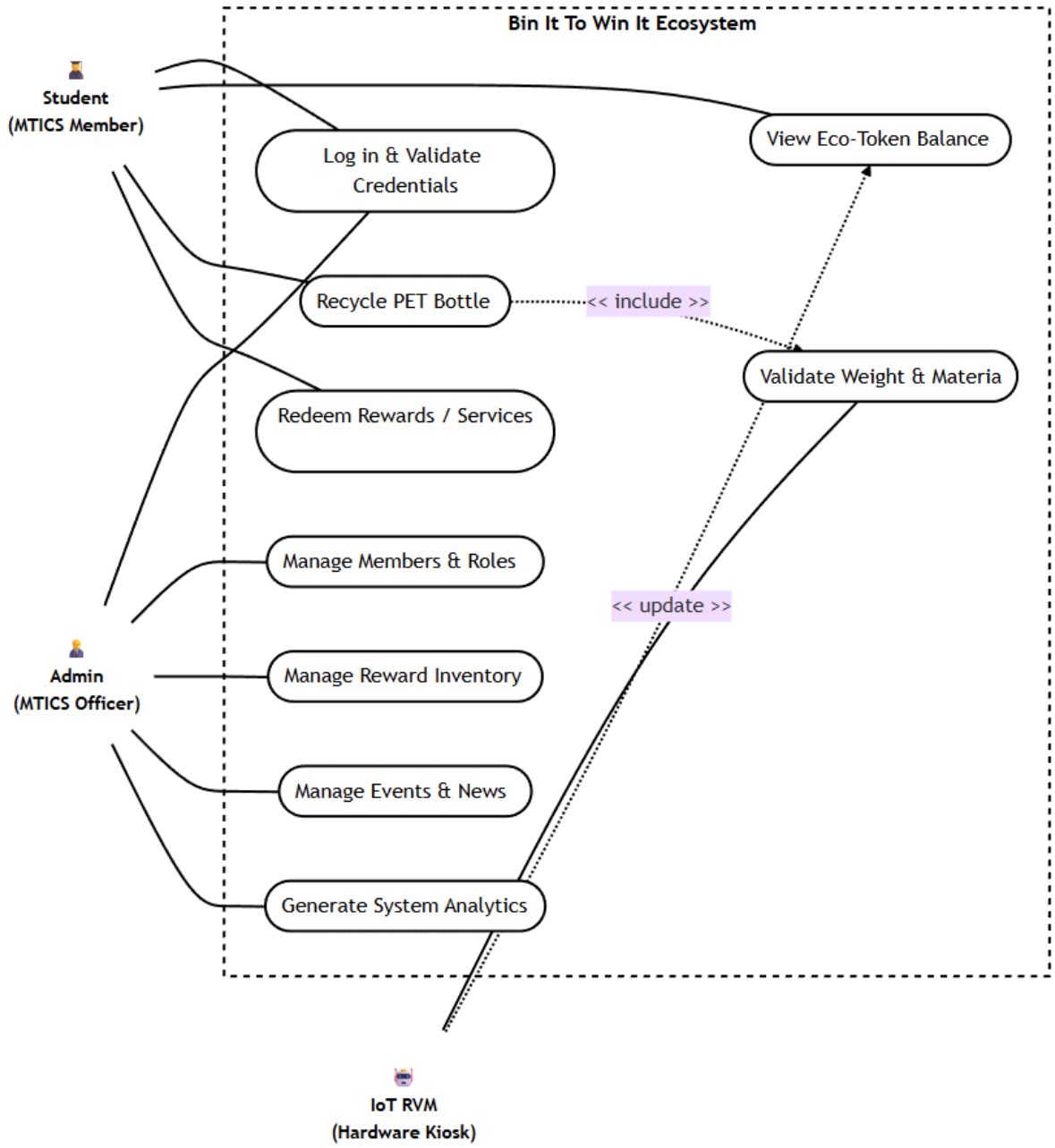

Figure 8. Use Case Diagram of Bin it To Win it: An MTICS Eco-nnect IoT-Integrated Reverse Vending & Web-Based Reward System

3.7. GANTT CHART

Gantt Chart for Bin it To Win it: An MTICS Eco-nnect IoT-Integrated Reverse Vending & Web-Based Reward System

Legend:

Name	Banania	Barredo	Caisip	Pili
Color				

Tasks	MONTH								
	DECEMBER		JANUARY				FEBRUARY		
Week	1	2	3	4	5	6	7	8	9
Planning									
Title Proposal									
Chapter 1: The Problem and Its Background									
Chapter 1 (Submission)									
Chapter 1 (Revision)									
Chapter 2: Review of Related Literature									
Chapter 1 and 2 (Submission)									
Chapter 3: Research Methodology									
Chapter 3 (Submission)									

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